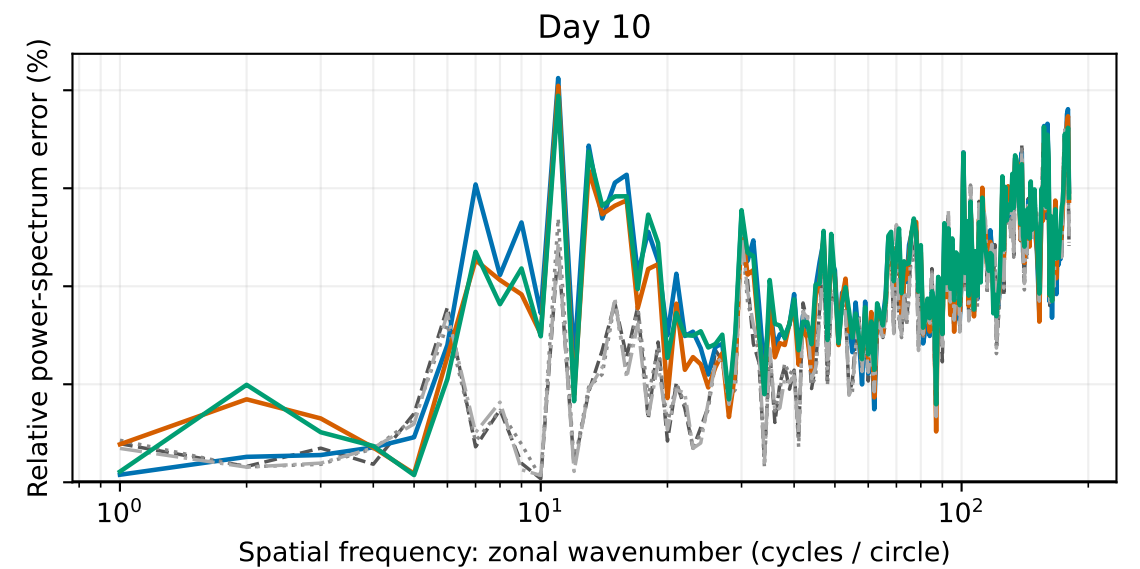
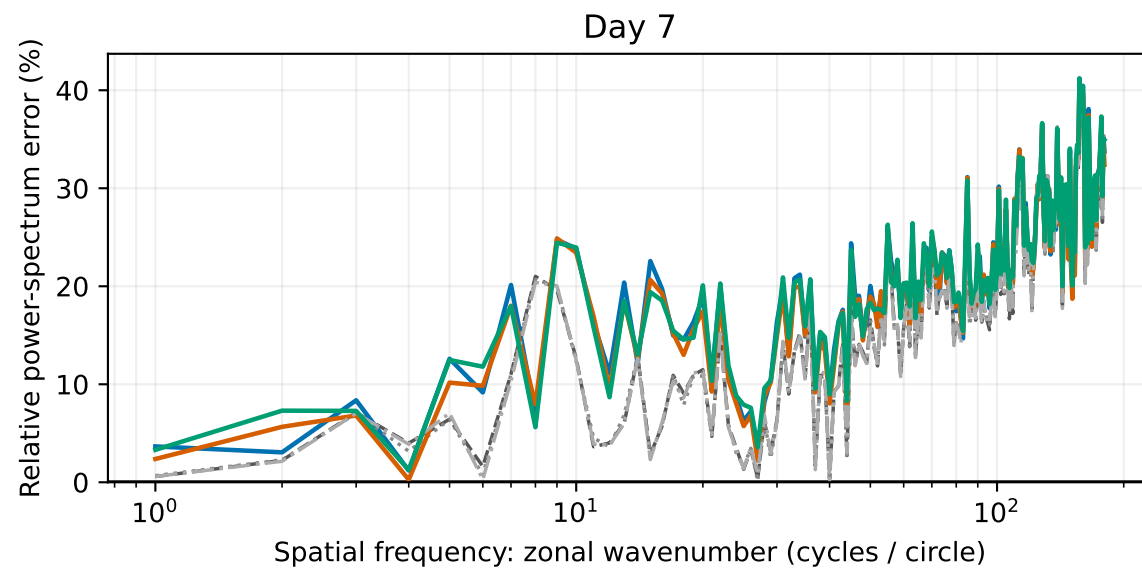
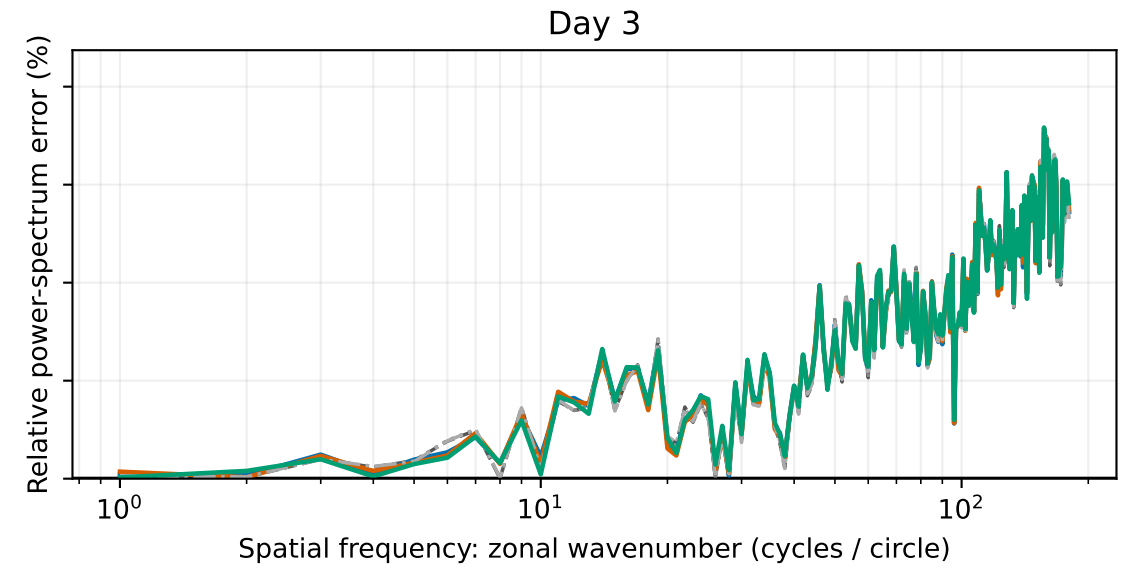
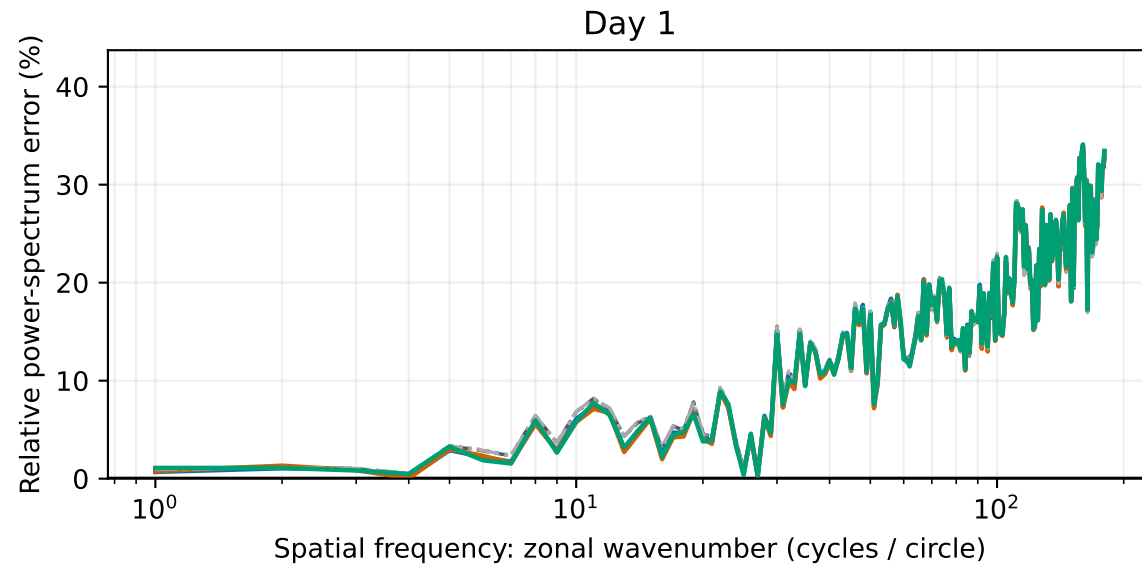


T2m: relative power-spectrum error versus spatial frequency

$100 \times |\text{forecast power} - \text{ERA5 power}| / \text{ERA5 power}$ · Lower is better · Zero is ideal

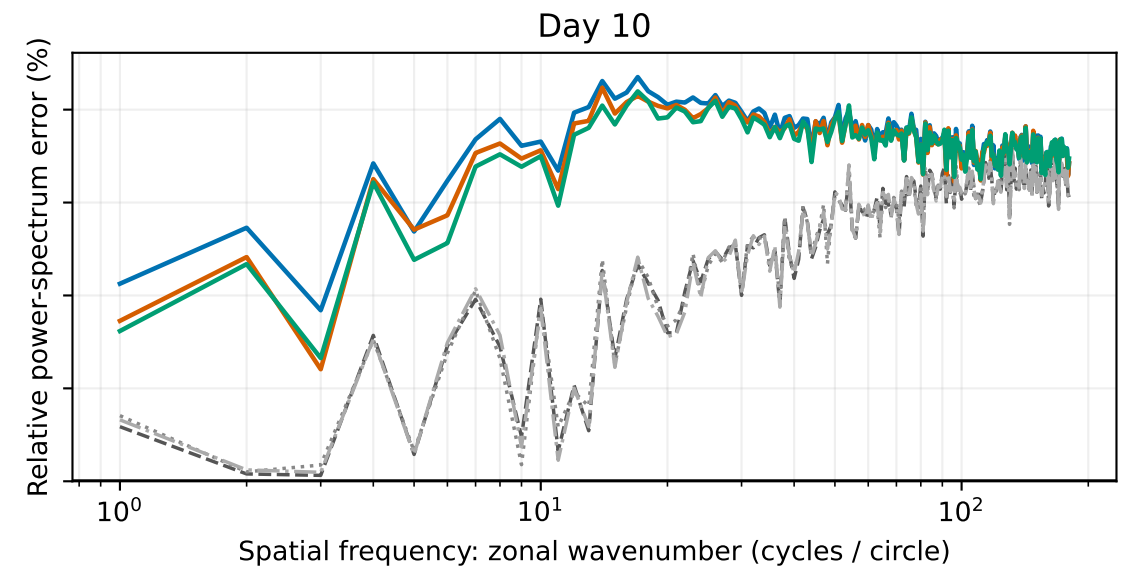
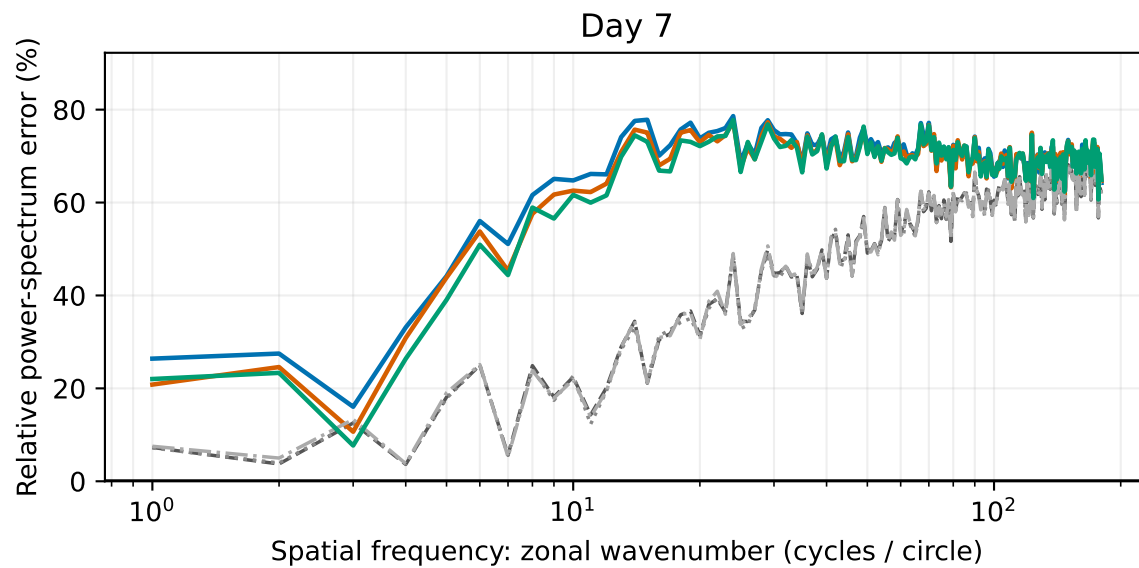
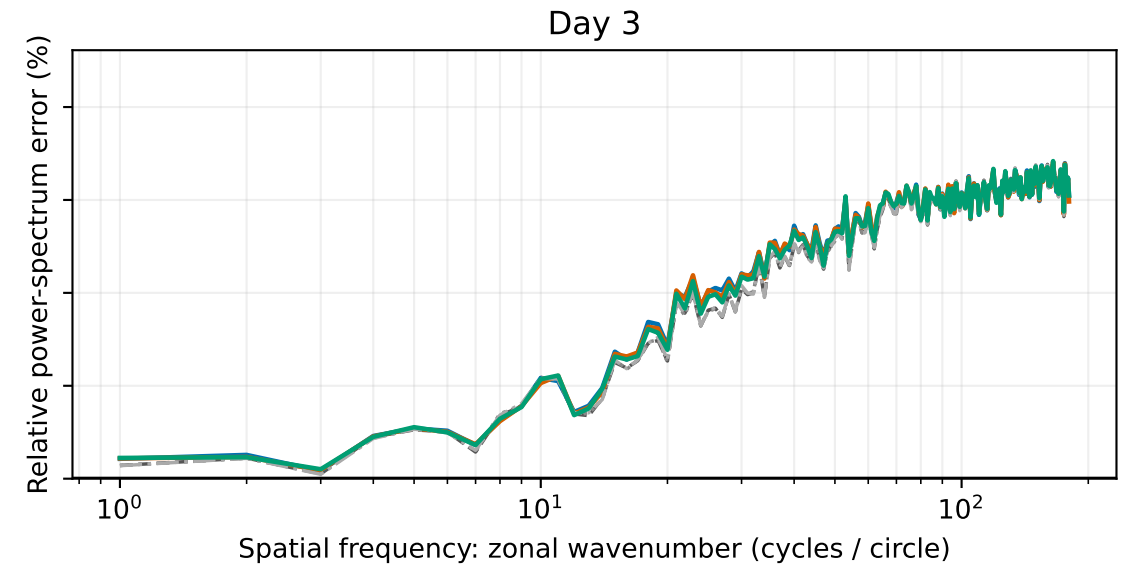
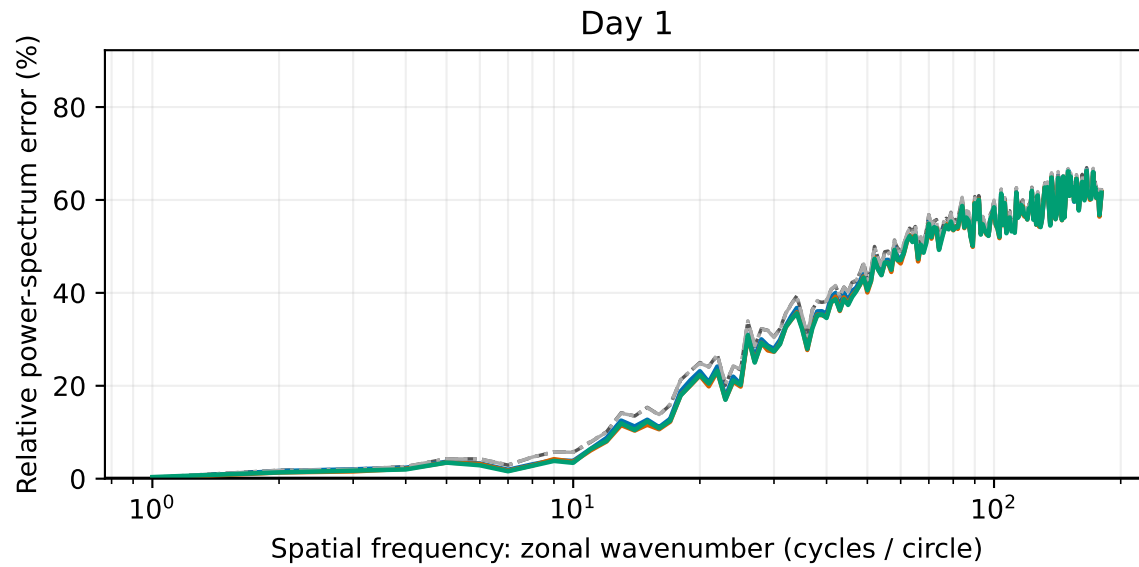


— ERA5: zero discrepancy ····· Original GC, evaluation 2 - - - Original GC, evaluation 3
 - - - Original GC, evaluation 1 — di32 · joint LR — di16 · optimizer study
 — di16 · faster Mamba LR

1° · Eight 2023 starts · 30–60° both hemispheres · Cosine latitude weights · Powers averaged before ratios
 Baseline repeatability unresolved: all three GC runs shown. Closer to zero means better power agreement.

U10: relative power-spectrum error versus spatial frequency

$100 \times |\text{forecast power} - \text{ERA5 power}| / \text{ERA5 power}$ · Lower is better · Zero is ideal

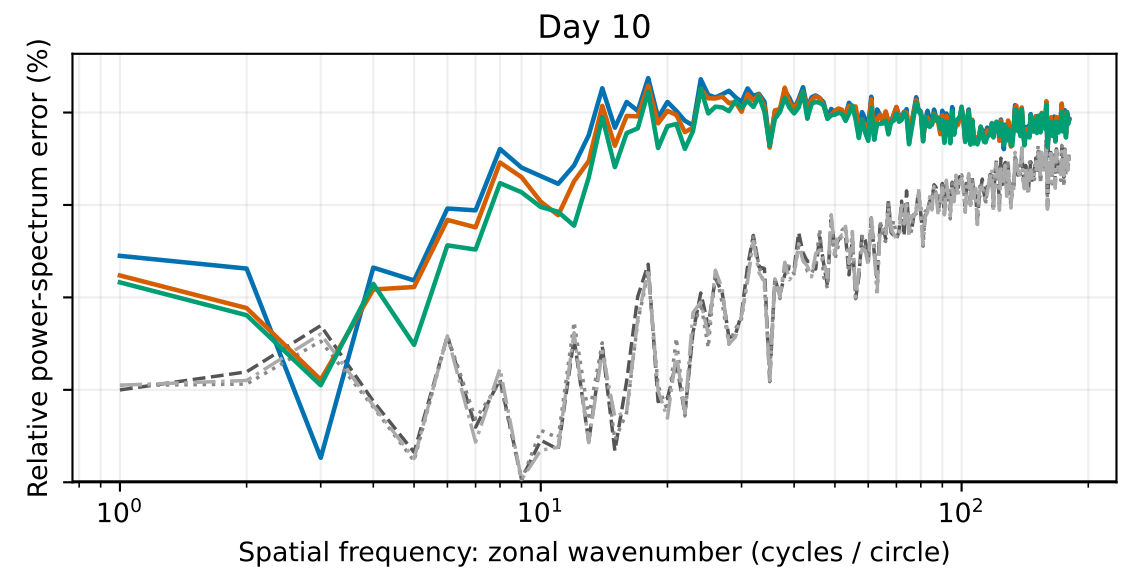
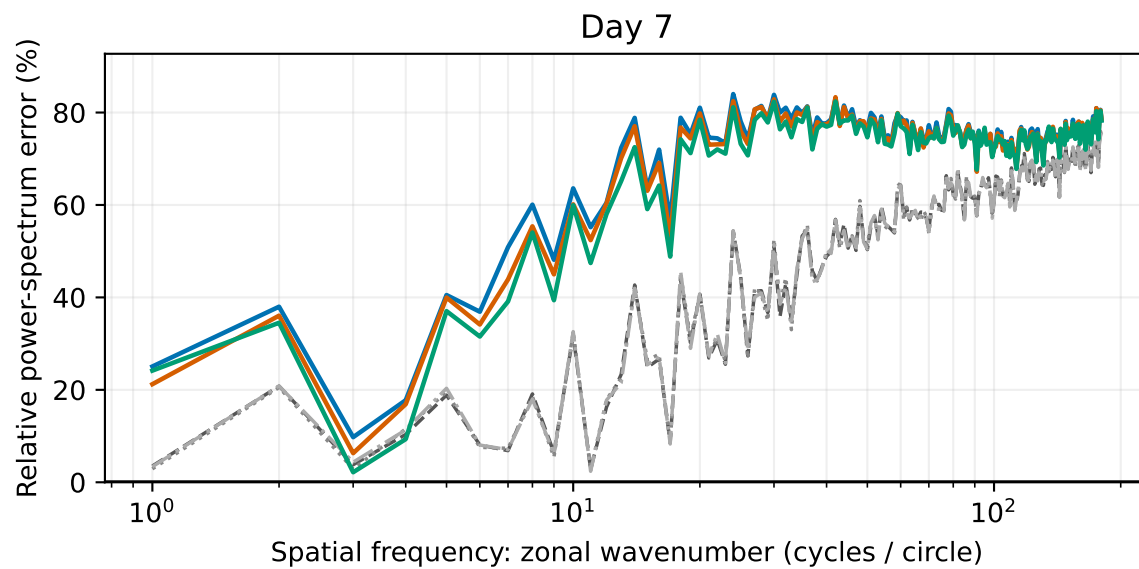
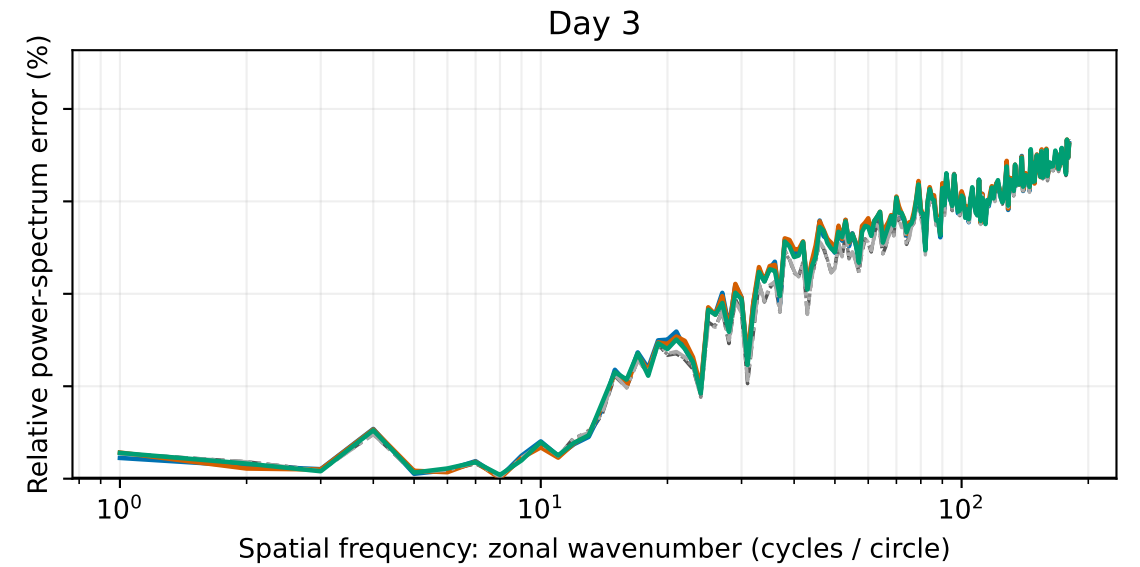
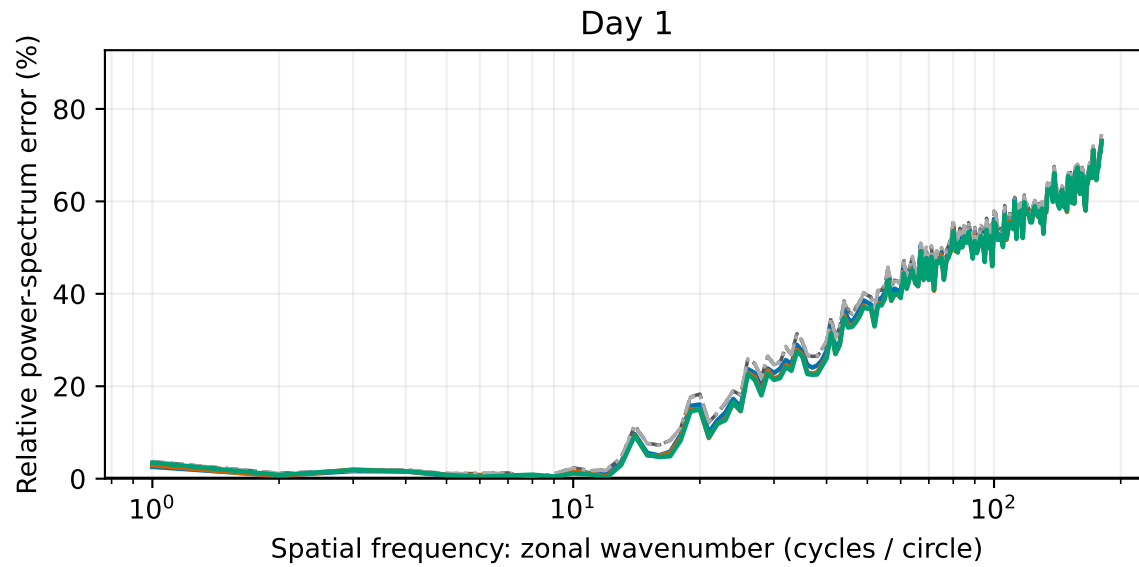


— ERA5: zero discrepancy Original GC, evaluation 2 - - - Original GC, evaluation 3
- - - Original GC, evaluation 1 — di32 · joint LR — di16 · optimizer study
— di16 · faster Mamba LR

1° · Eight 2023 starts · 30–60° both hemispheres · Cosine latitude weights · Powers averaged before ratios
Baseline repeatability unresolved: all three GC runs shown. Closer to zero means better power agreement.

V10: relative power-spectrum error versus spatial frequency

$100 \times |\text{forecast power} - \text{ERA5 power}| / \text{ERA5 power}$ · Lower is better · Zero is ideal

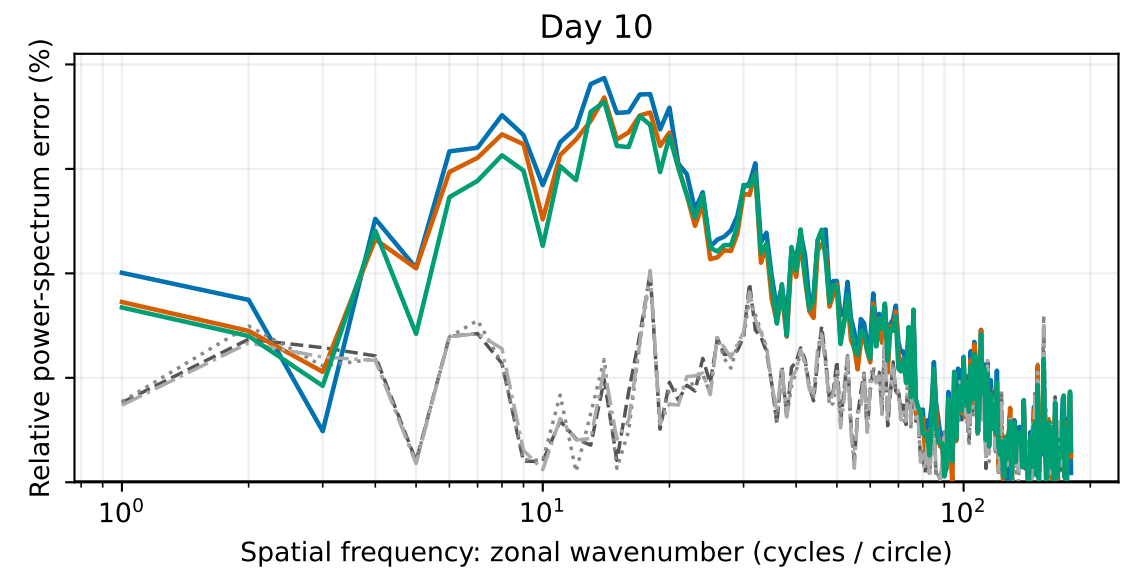
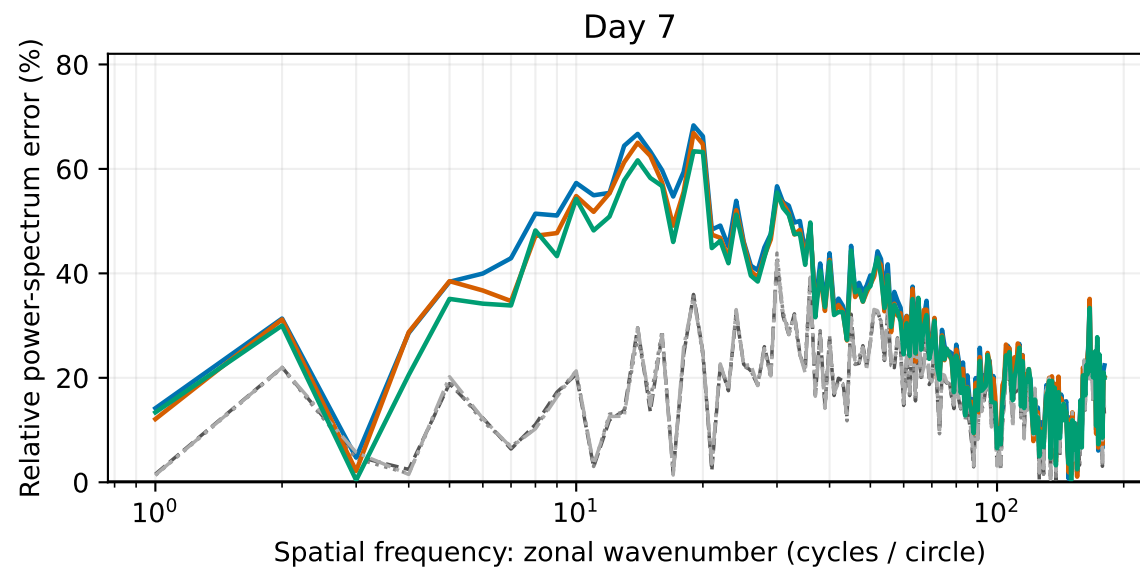
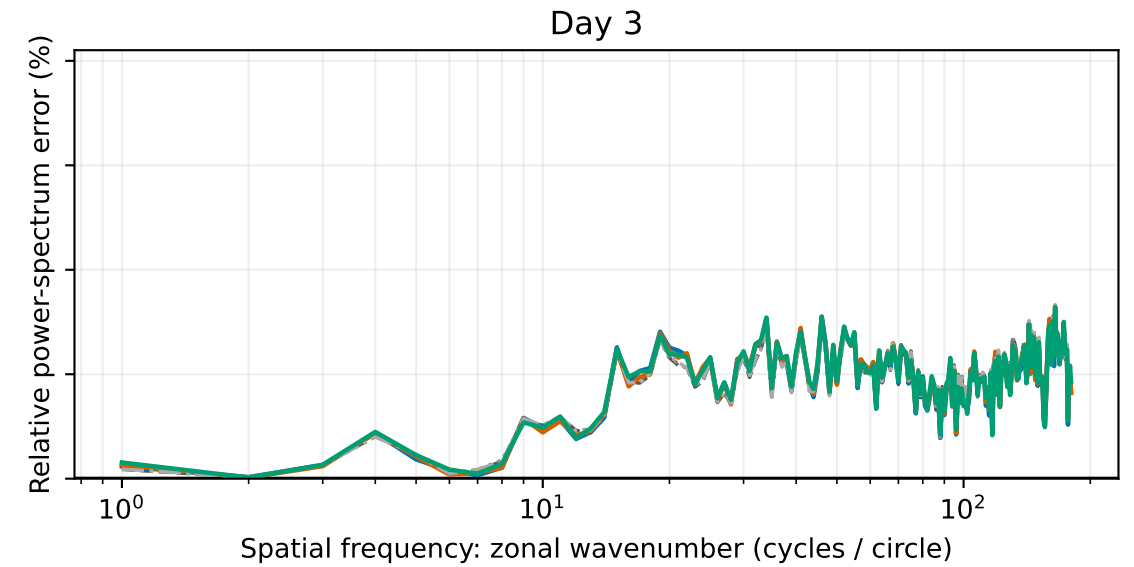
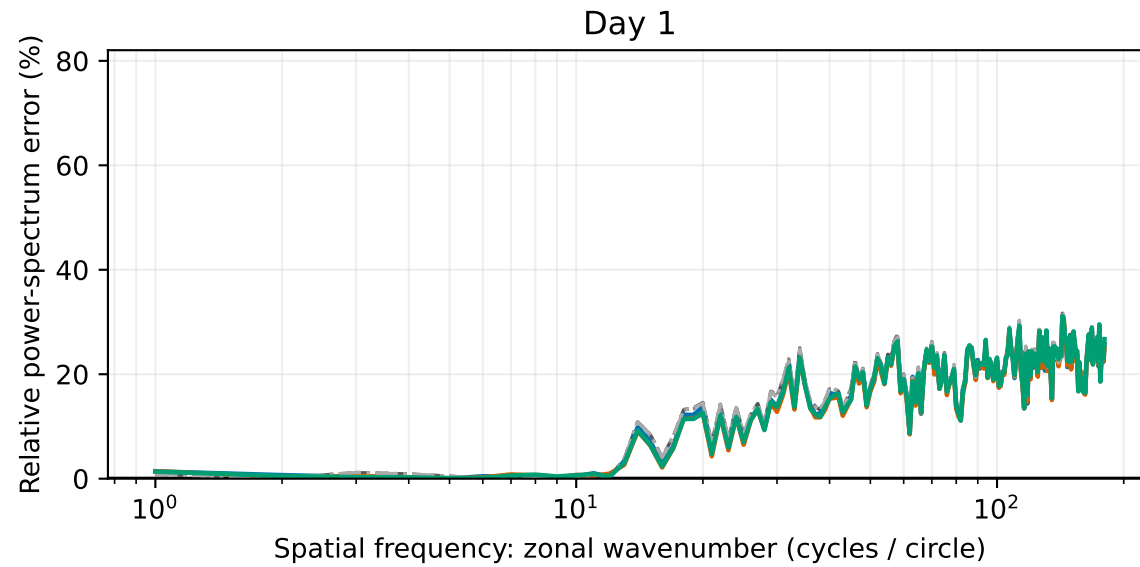


— ERA5: zero discrepancy Original GC, evaluation 2 - - - Original GC, evaluation 3
- - - Original GC, evaluation 1 — di32 · joint LR — di16 · optimizer study
— di16 · faster Mamba LR

1° · Eight 2023 starts · 30–60° both hemispheres · Cosine latitude weights · Powers averaged before ratios
Baseline repeatability unresolved: all three GC runs shown. Closer to zero means better power agreement.

MSLP: relative power-spectrum error versus spatial frequency

$100 \times |\text{forecast power} - \text{ERA5 power}| / \text{ERA5 power}$ · Lower is better · Zero is ideal

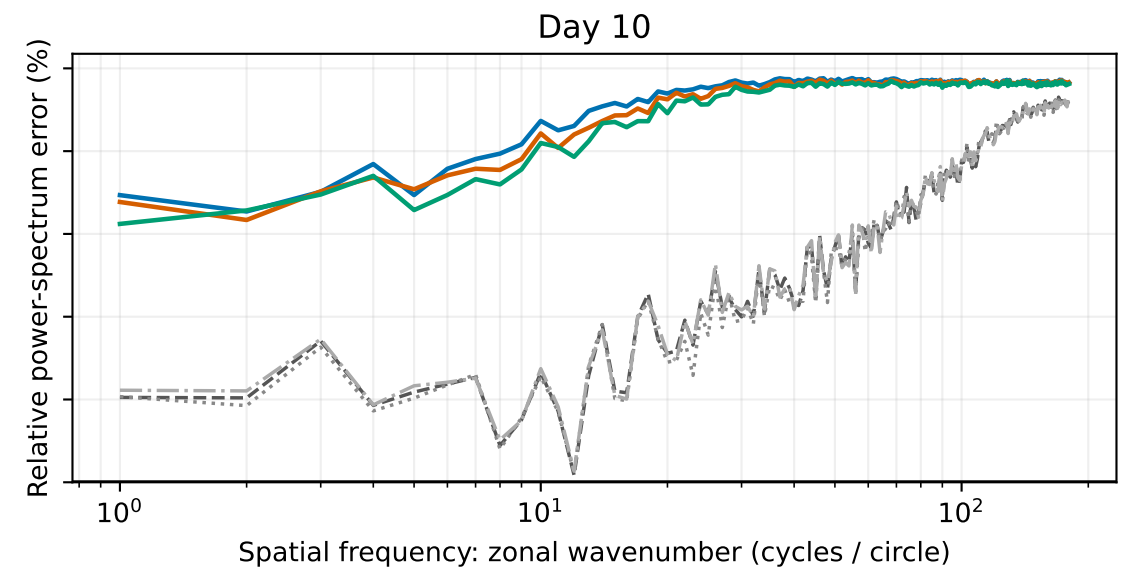
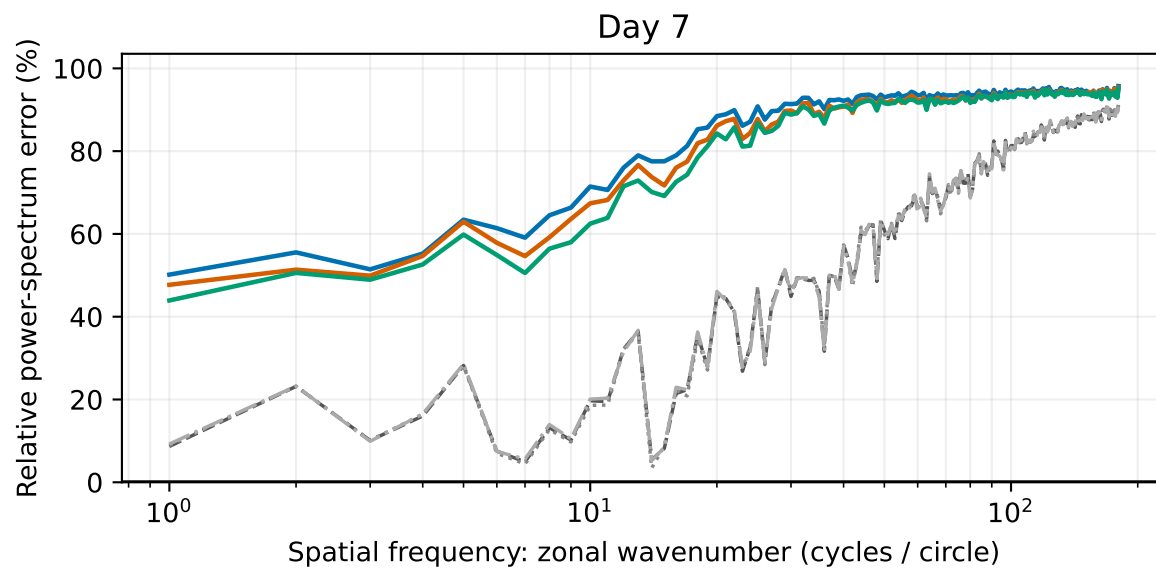
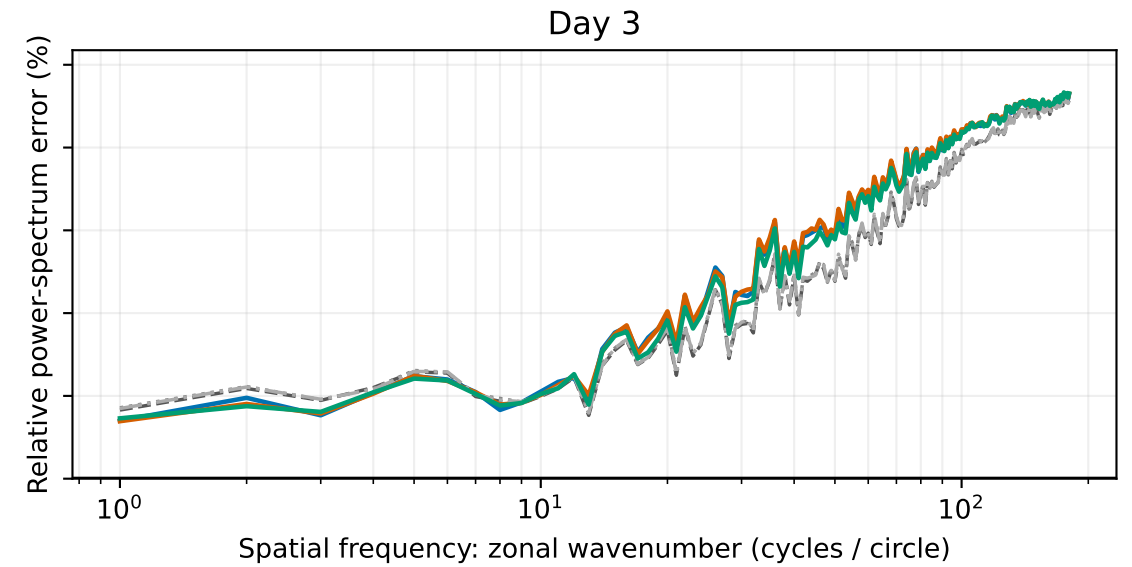
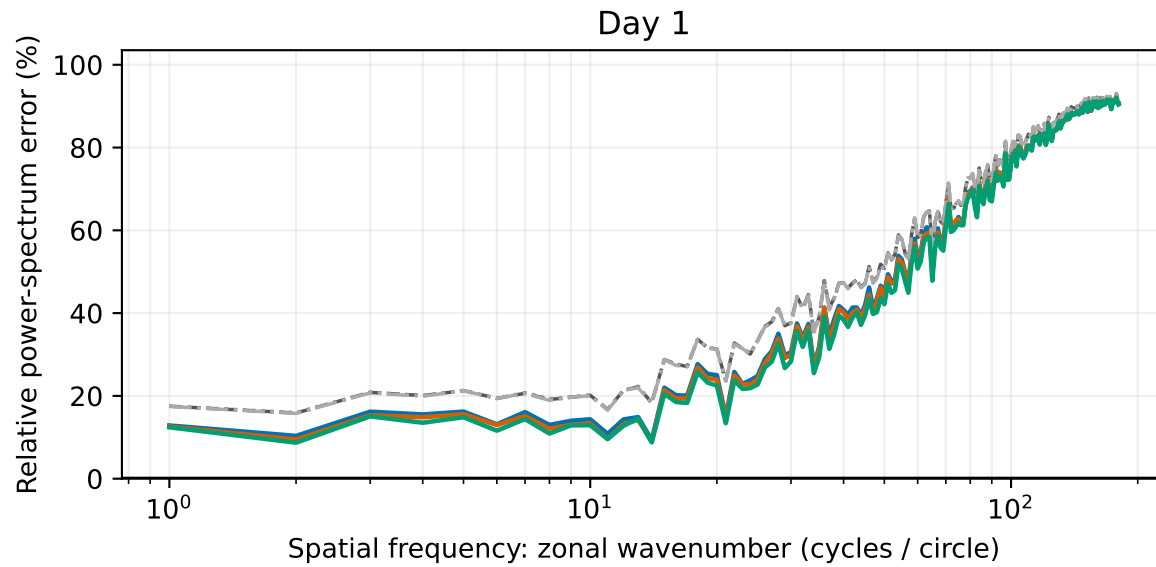


— ERA5: zero discrepancy ····· Original GC, evaluation 2 - - - Original GC, evaluation 3
- - - Original GC, evaluation 1 — di32 · joint LR — di16 · optimizer study
— di16 · faster Mamba LR

1° · Eight 2023 starts · 30–60° both hemispheres · Cosine latitude weights · Powers averaged before ratios
Baseline repeatability unresolved: all three GC runs shown. Closer to zero means better power agreement.

TP6: relative power-spectrum error versus spatial frequency

$100 \times |\text{forecast power} - \text{ERA5 power}| / \text{ERA5 power}$ · Lower is better · Zero is ideal

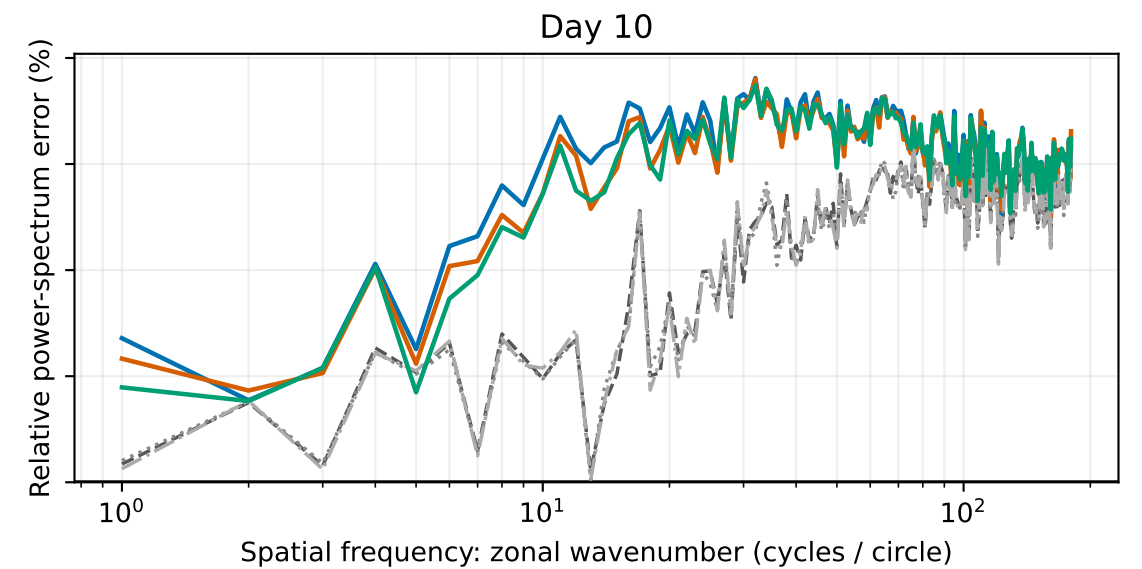
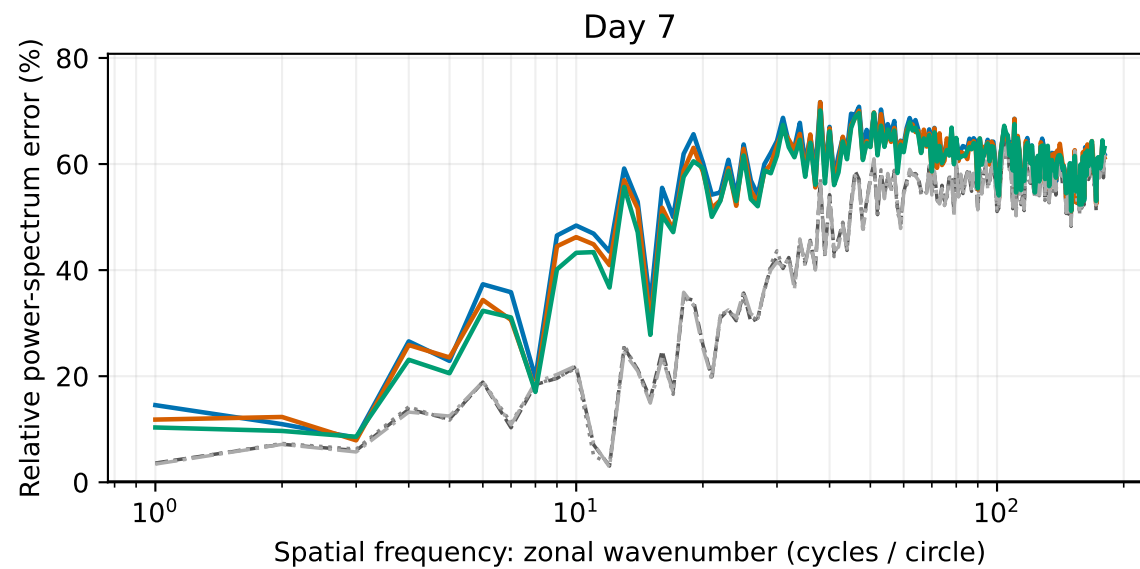
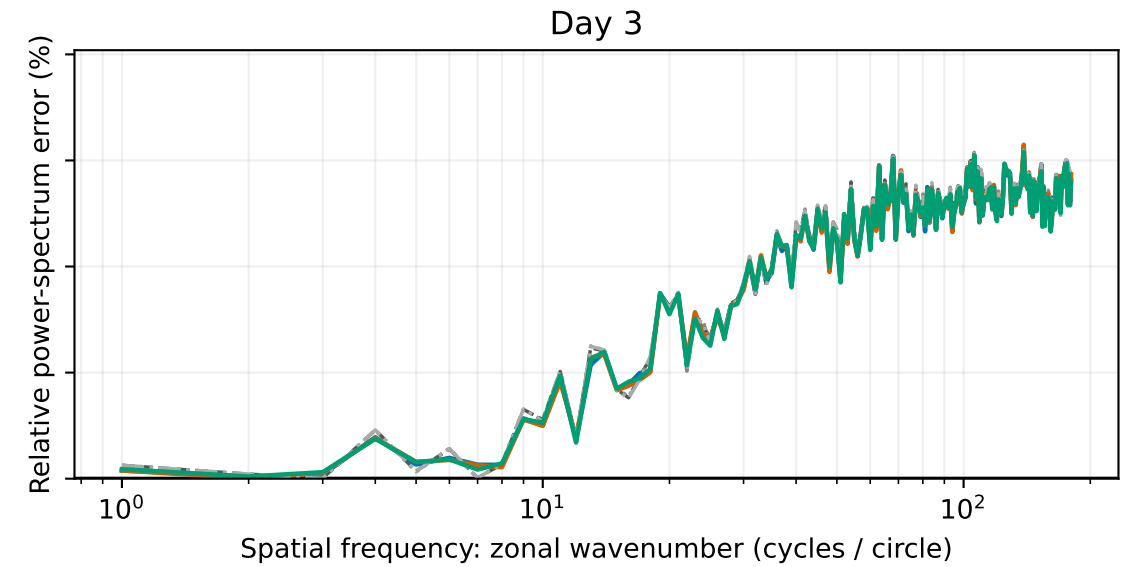
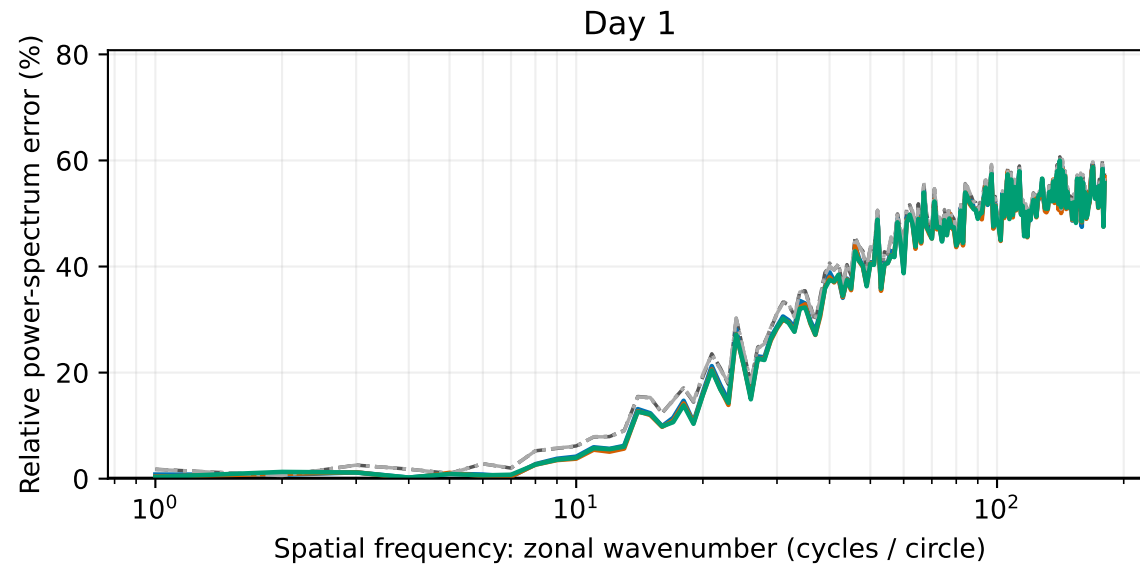


— ERA5: zero discrepancy ····· Original GC, evaluation 2 - - - Original GC, evaluation 3
 - - - Original GC, evaluation 1 — di32 · joint LR — di16 · optimizer study
 — di16 · faster Mamba LR

1° · Eight 2023 starts · 30–60° both hemispheres · Cosine latitude weights · Powers averaged before ratios
 Baseline repeatability unresolved: all three GC runs shown. Closer to zero means better power agreement.

T850: relative power-spectrum error versus spatial frequency

$100 \times |\text{forecast power} - \text{ERA5 power}| / \text{ERA5 power}$ · Lower is better · Zero is ideal

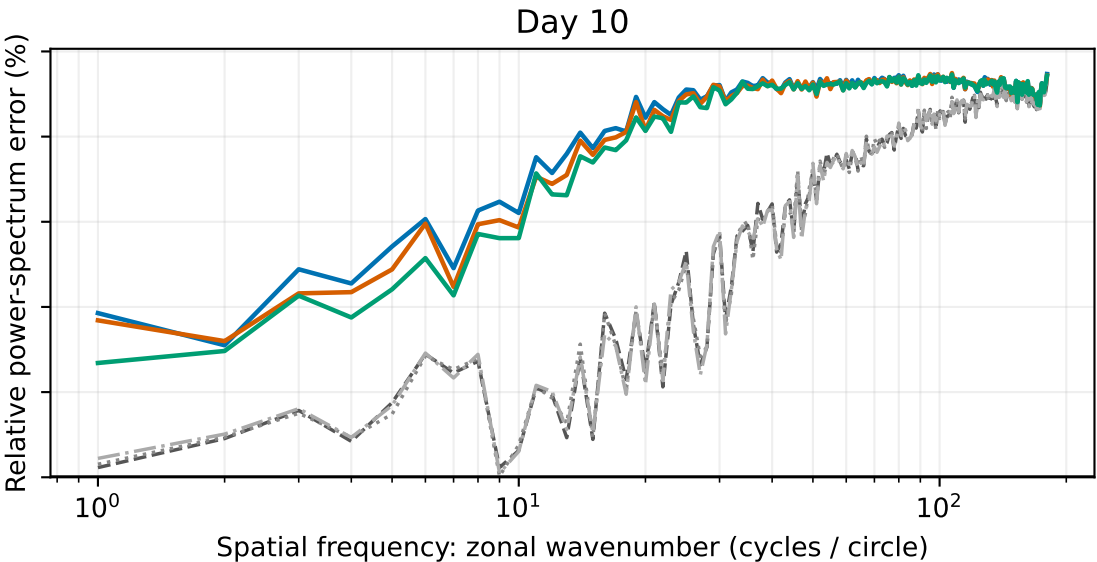
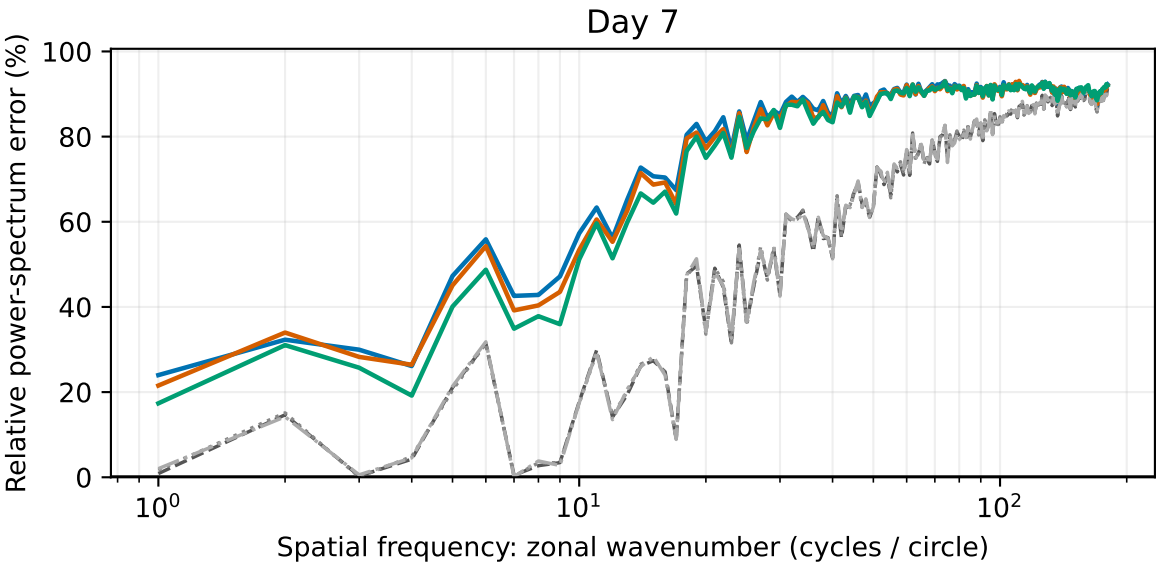
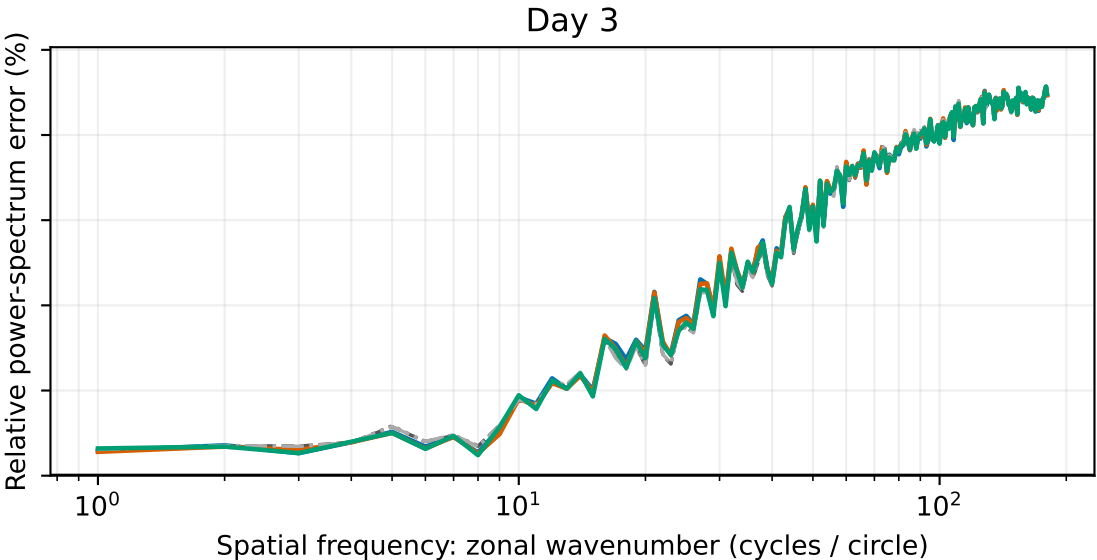
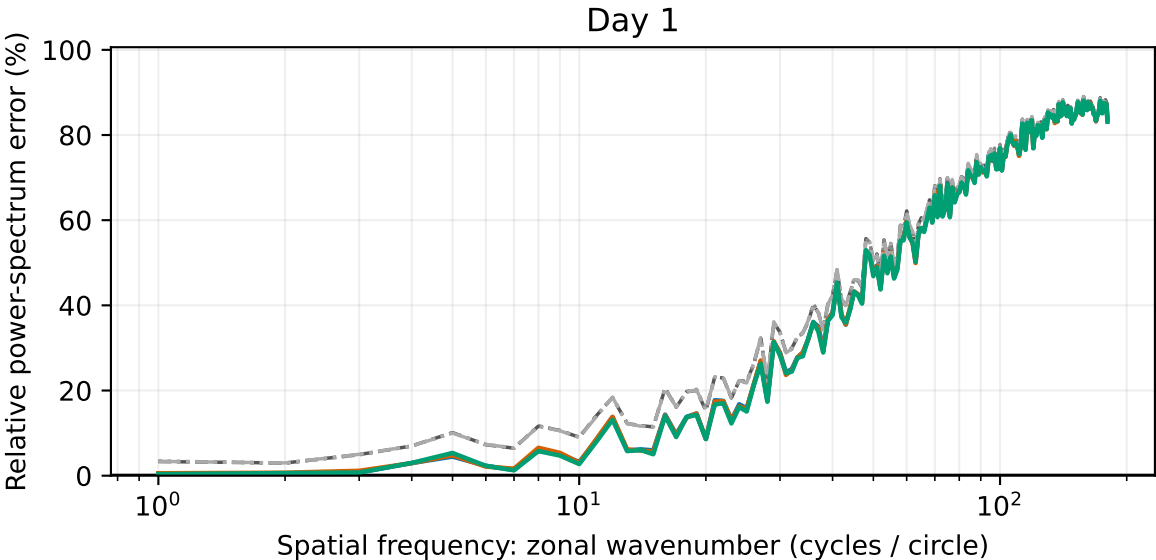


— ERA5: zero discrepancy	····· Original GC, evaluation 2	— Original GC, evaluation 3
- - - Original GC, evaluation 1	— di32 · joint LR	— di16 · optimizer study
— di16 · faster Mamba LR		

1° · Eight 2023 starts · 30–60° both hemispheres · Cosine latitude weights · Powers averaged before ratios
Baseline repeatability unresolved: all three GC runs shown. Closer to zero means better power agreement.

Q700: relative power-spectrum error versus spatial frequency

$100 \times |\text{forecast power} - \text{ERA5 power}| / \text{ERA5 power}$ · Lower is better · Zero is ideal

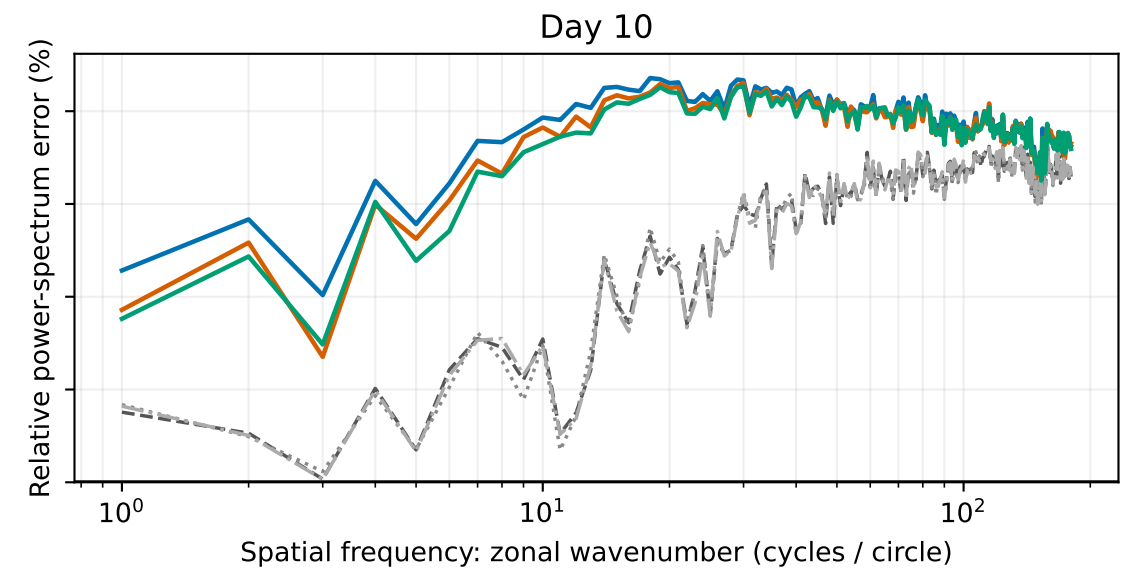
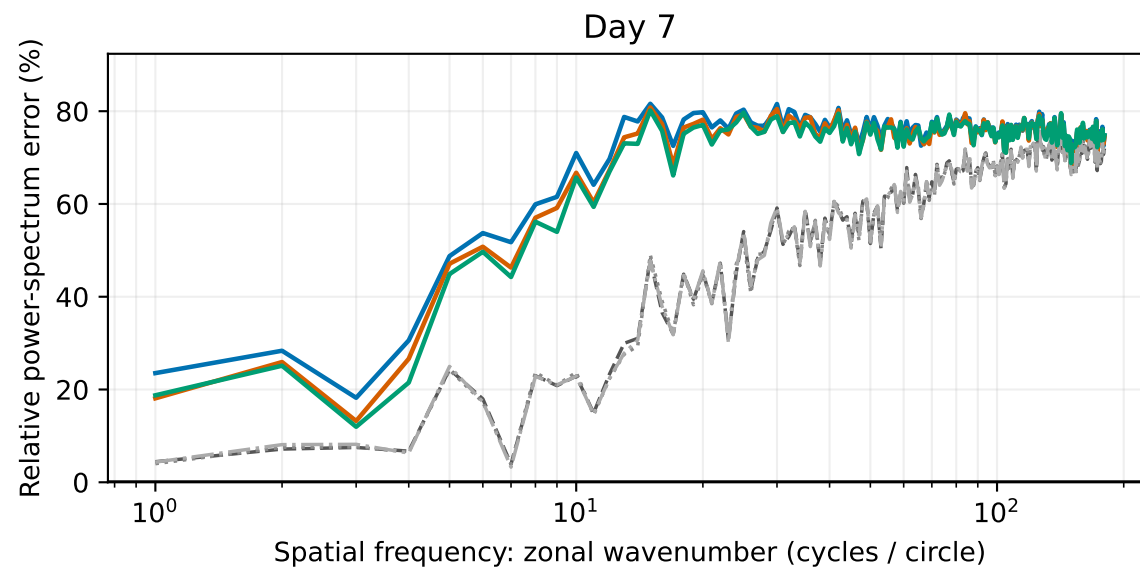
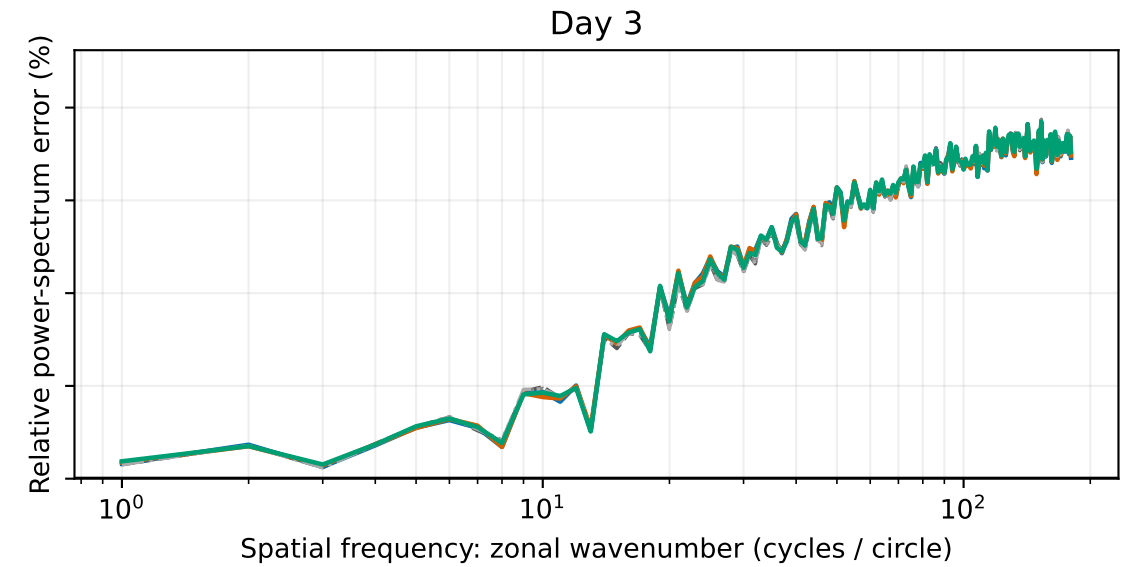
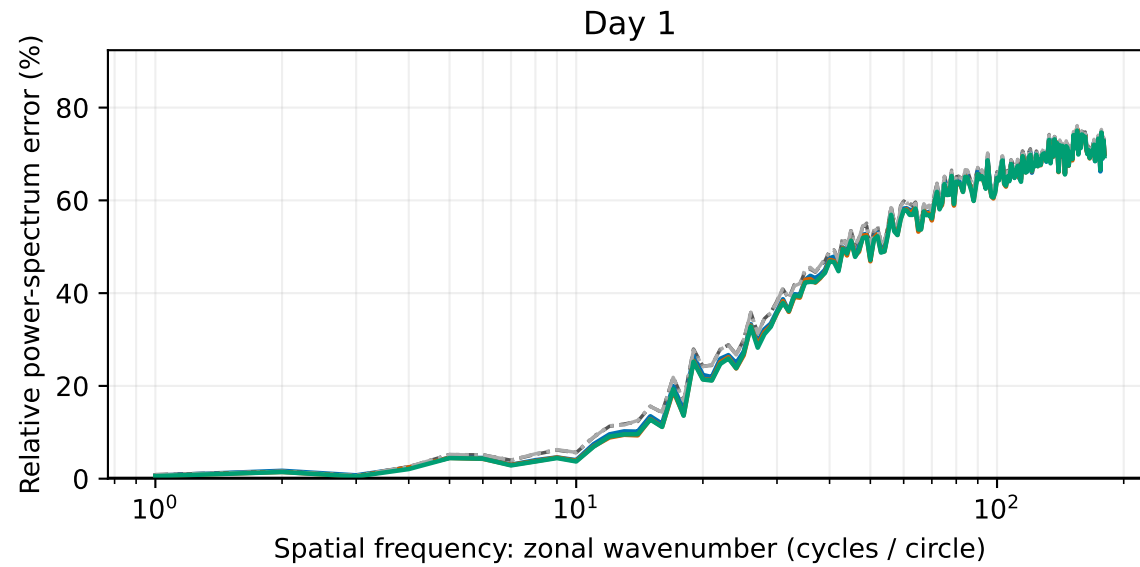


— ERA5: zero discrepancy Original GC, evaluation 2 - - - Original GC, evaluation 3
- - - Original GC, evaluation 1 — di32 · joint LR — di16 · optimizer study
— di16 · faster Mamba LR

1° · Eight 2023 starts · 30–60° both hemispheres · Cosine latitude weights · Powers averaged before ratios
Baseline repeatability unresolved: all three GC runs shown. Closer to zero means better power agreement.

U850: relative power-spectrum error versus spatial frequency

$100 \times |\text{forecast power} - \text{ERA5 power}| / \text{ERA5 power}$ · Lower is better · Zero is ideal

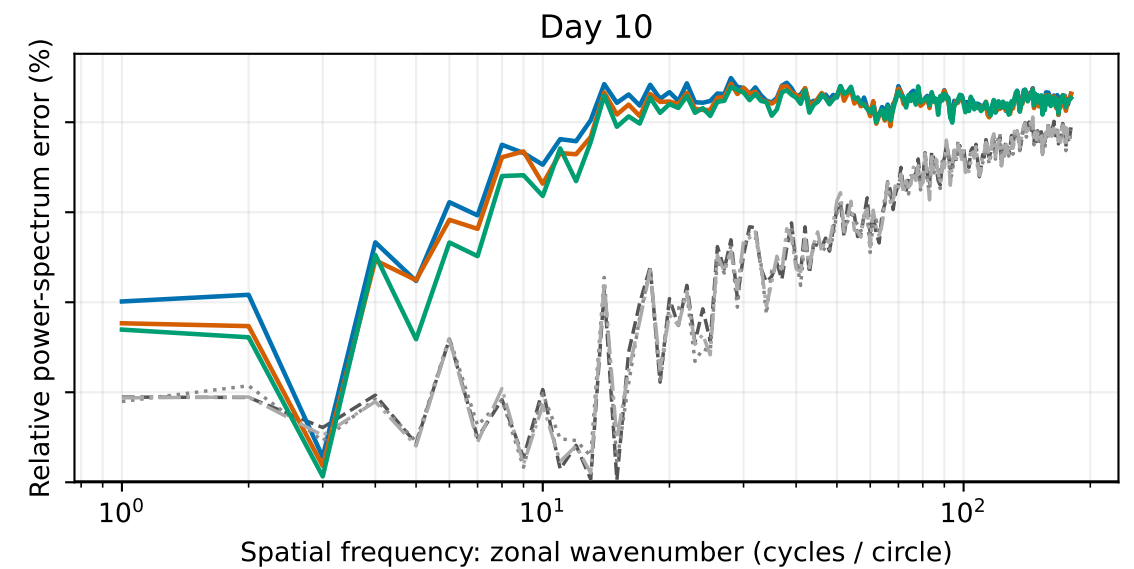
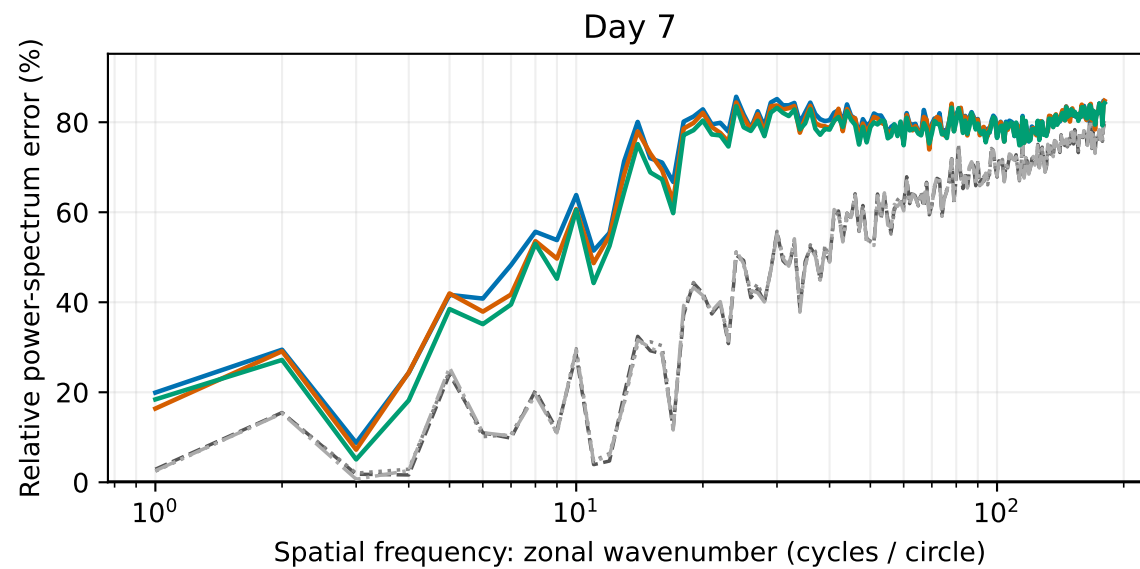
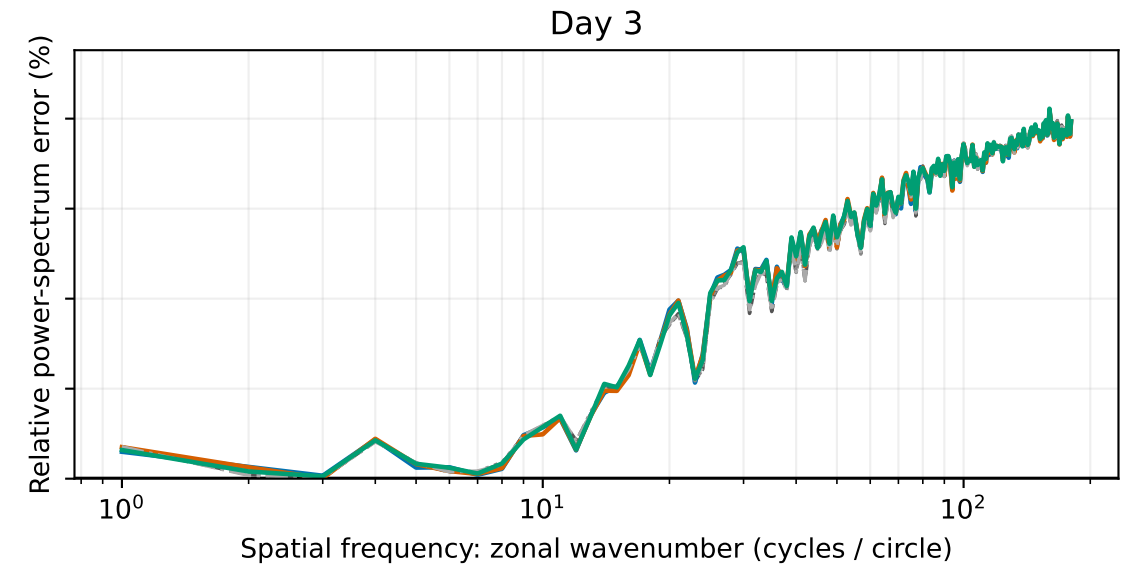
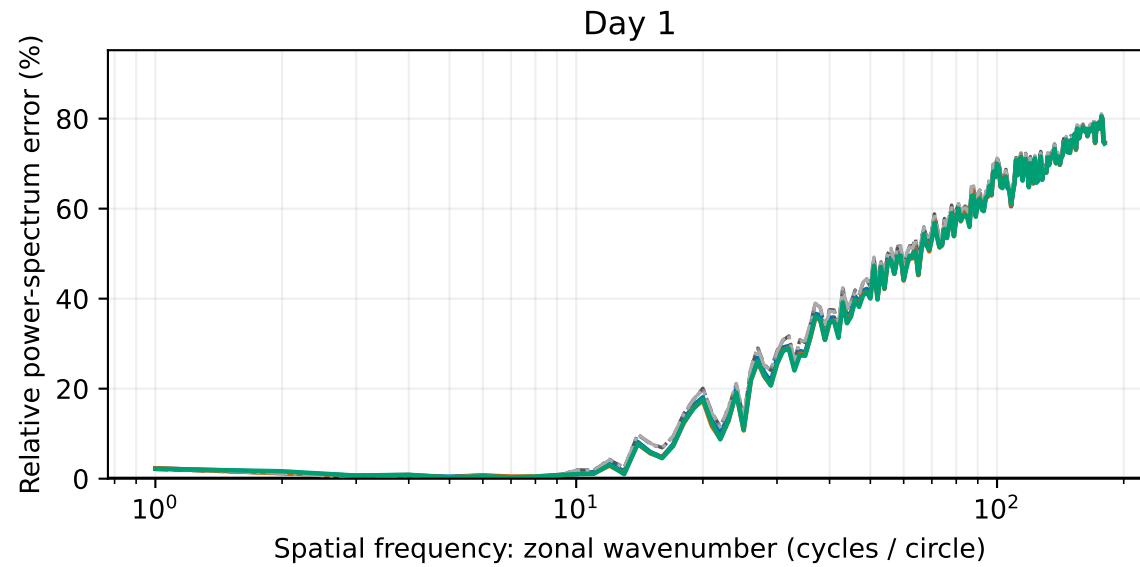


— ERA5: zero discrepancy ····· Original GC, evaluation 2 - - - Original GC, evaluation 3
- - - Original GC, evaluation 1 — di32 · joint LR — di16 · optimizer study
— di16 · faster Mamba LR

1° · Eight 2023 starts · 30–60° both hemispheres · Cosine latitude weights · Powers averaged before ratios
Baseline repeatability unresolved: all three GC runs shown. Closer to zero means better power agreement.

V850: relative power-spectrum error versus spatial frequency

$100 \times |\text{forecast power} - \text{ERA5 power}| / \text{ERA5 power}$ · Lower is better · Zero is ideal

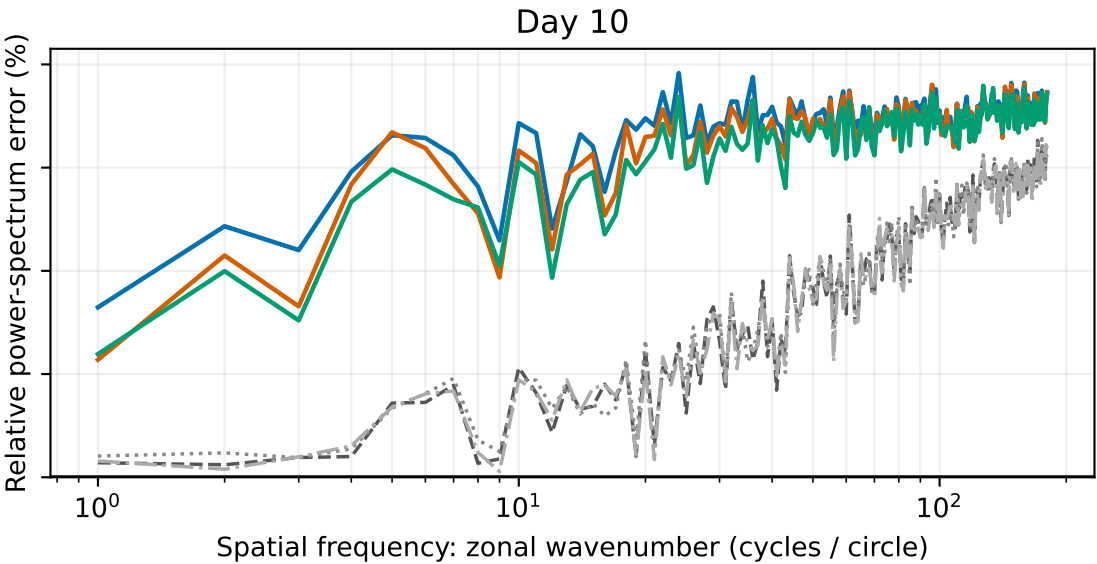
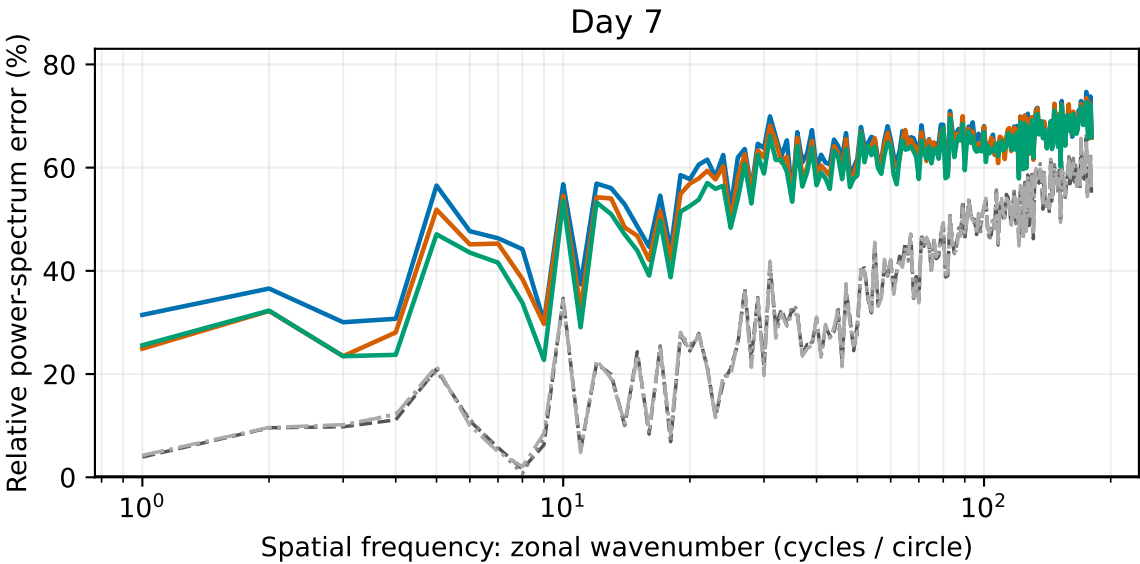
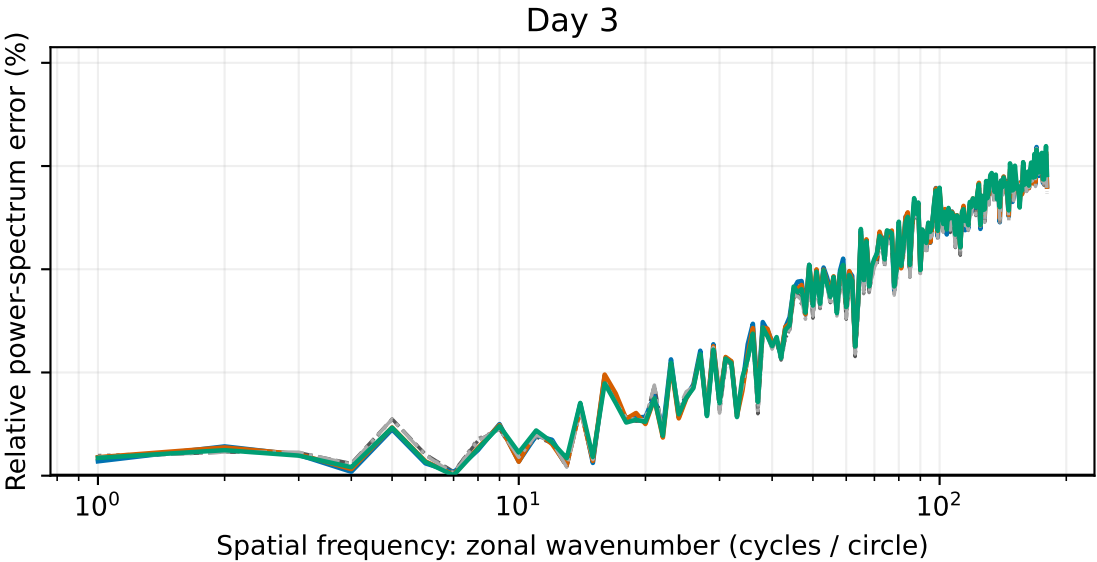
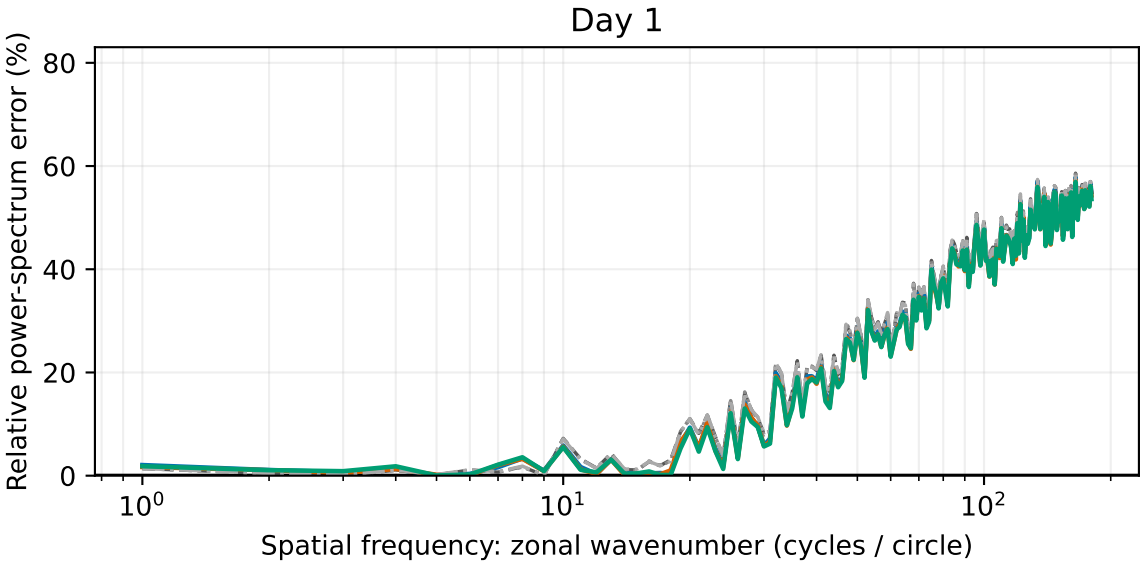


— ERA5: zero discrepancy ····· Original GC, evaluation 2 - - - Original GC, evaluation 3
 - - - Original GC, evaluation 1 — di32 · joint LR — di16 · optimizer study
 — di16 · faster Mamba LR

1° · Eight 2023 starts · 30–60° both hemispheres · Cosine latitude weights · Powers averaged before ratios
 Baseline repeatability unresolved: all three GC runs shown. Closer to zero means better power agreement.

WS10: relative power-spectrum error versus spatial frequency

$100 \times |\text{forecast power} - \text{ERA5 power}| / \text{ERA5 power}$ · Lower is better · Zero is ideal

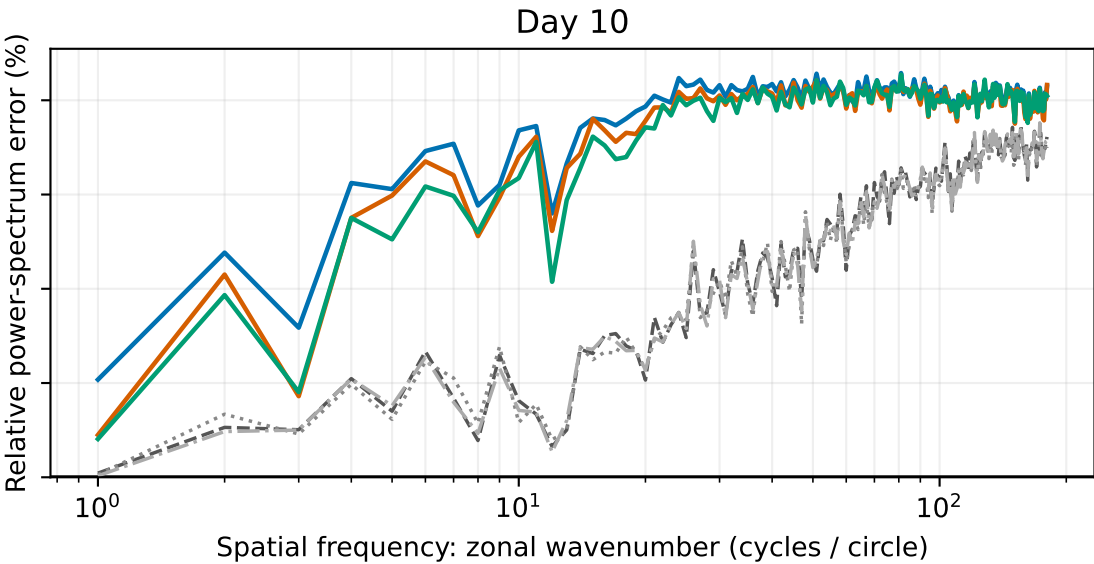
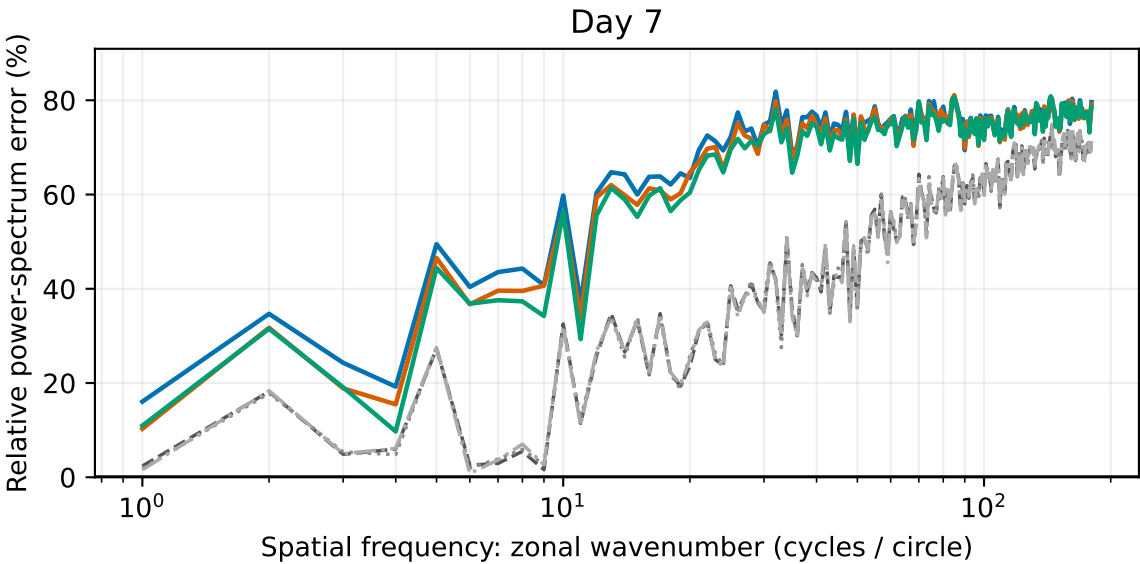
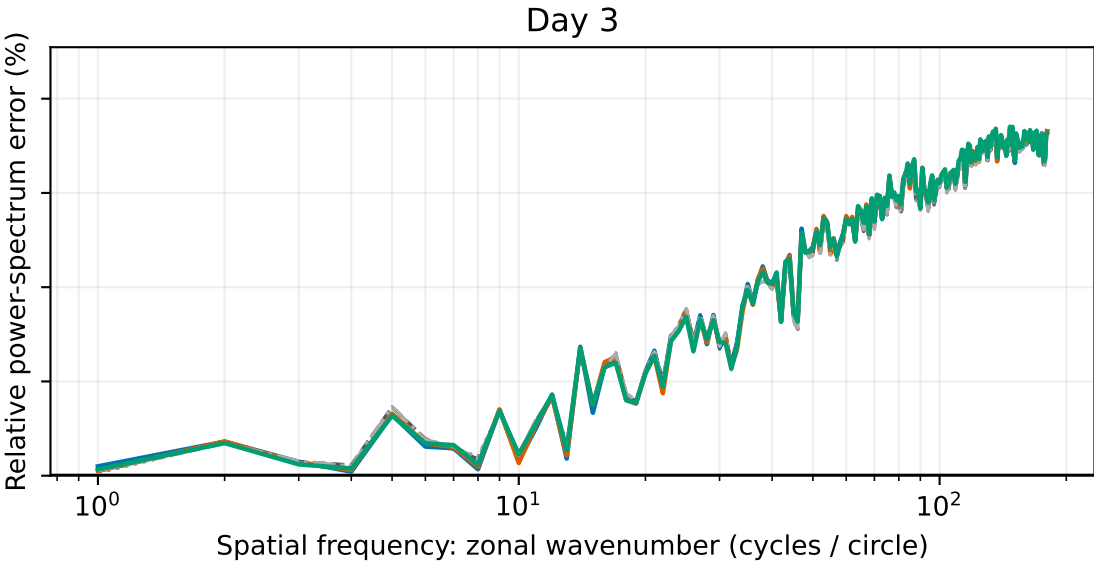
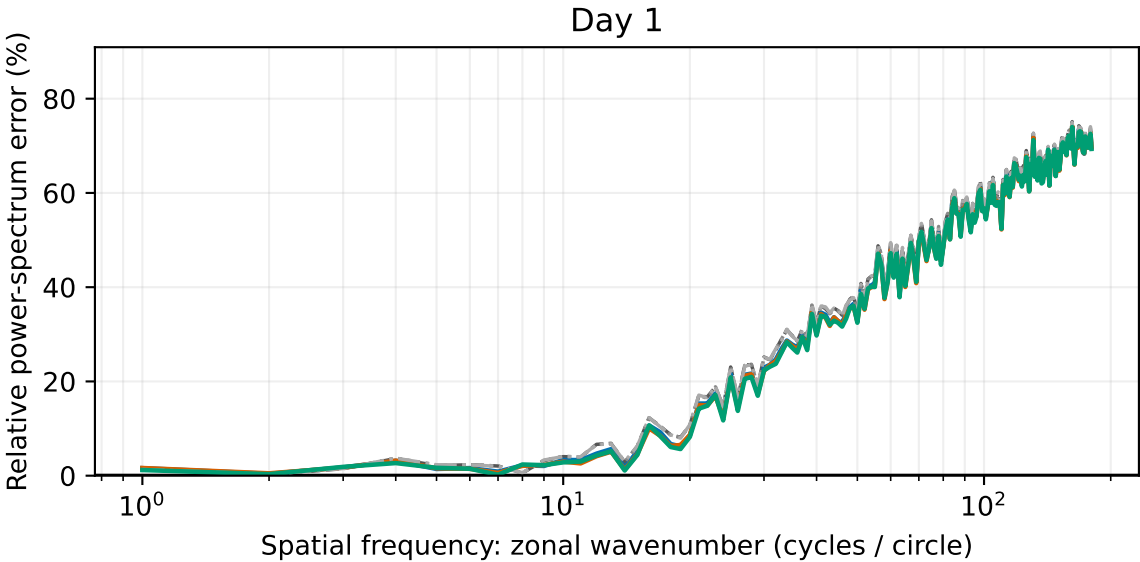


— ERA5: zero discrepancy Original GC, evaluation 2 - - - Original GC, evaluation 3
- - - Original GC, evaluation 1 — di32 · joint LR — di16 · optimizer study
— di16 · faster Mamba LR

1° · Eight 2023 starts · 30–60° both hemispheres · Cosine latitude weights · Powers averaged before ratios
Baseline repeatability unresolved: all three GC runs shown. Closer to zero means better power agreement.

WS850: relative power-spectrum error versus spatial frequency

$100 \times |\text{forecast power} - \text{ERA5 power}| / \text{ERA5 power}$ · Lower is better · Zero is ideal



- ERA5: zero discrepancy

--- Original GC, evaluation 1

— di16 · faster Mamba LR
- Original GC, evaluation 2

— di32 · joint LR

— di16 · optimizer study
- Original GC, evaluation 3

1° · Eight 2023 starts · 30–60° both hemispheres · Cosine latitude weights · Powers averaged before ratios
Baseline repeatability unresolved: all three GC runs shown. Closer to zero means better power agreement.